Seasonality Analysis for agricultural products

As of 2010, California is the top agricultural state with $37.5 Billion in Gross Cash Receipts, which led authority to national largest agricultural producer and exporter. As consider the nature of agricultural products, transport process should be simple and expeditious. Therefore, a large share of agricultural commodity is assigned to freight trucks which directly affect on California freight corridors.

An interesting feature of agricultural products is seasonality because the harvest period varies by agricultural products. Seasonality analysis for agricultural products is far more complex than the one performed for fuel and other similar commodities, and this can be explained by three different reasons: The first reason is that the production is spread throughout the State, and not concentrated in known refineries. The second reason is that agricultural products consist of various crop and livestock commodities, whose cultivating and harvesting seasons vary (See Appendix 2). The third reason is the lack of consistent survey or census data for the whole set of products.

1 Methodology

Due to the heterogeneity of the agricultural commodities, it was necessary to estimate the production and the corresponding seasonality (harvesting period) by agricultural product in order to compute an average distribution of the agricultural commodities that originate in each Freight Analysis Zone (FAZ). To achieve such objective, three pieces of information for each FAZ in California requires: Crop areas, Crop yields and harvesting periods.

The basic idea is to divide the year into analysis periods (days, months and/or seasons) and compute the total production by crop for each one of the FAZs. By adding all the crops for a certain FAZ in a period (day, month and/or season), and dividing it by the total production of this FAZ in a year, the given period's percentage share of the total annual production can be calculated.

In other words, this methodology computes the distribution of the production through the year and for the analysis of any period of interest. Although it estimates agricultural production’s seasonality rather than transportation’s seasonality, it can be considered that transportation's seasonality are identical to production's seasonality because transportation happens whenever harvesting happens.

2 Data Sources

As it was expected, the level of data availability, format and units (i.e., crop specific units) used by official institutions were not homogeneous for the crops in California. Consequently, seasonality analysis confronted the additional problem of data compatibilization, which was done manually.

Three major data sources formed the primary basis for this analysis: CropScape (from USDA), USDA census and surveys data center (QuickStats) and USDA California report for 2010, which will be discussed in 2.1-2.2.

It is noteworthy however, that not only official data sources were used, as there was information that was not available in such sources. It was recorded, however, the source of each piece of information used on seasonality computation, and the recreation of the number presented in this report would not be a challenge.

2.1 CropScape (Crop areas)

The geographic raster layers existing in CropScape present an analysis using satellite and automated classification (with field validation) images taken periodically[1].

One of the greatest advantages using CropScape data can assure all the area of our proposed FAZ for California and the corresponding information on crops. The minor limitation may contain some errors that are intrinsic to the procedure of analyzing such images.

2.1.1 Some examples and statistics

Just in order to better illustrate the information that was obtained through CropScape, it was generated a map of Fresno County color-coded for all crops existing in such region, which is depicted in Figure 2.1.

[1] The technical details of this satellite imaging processing are not particularly relevant for the scope of the CSFFM, but further details can be found at http://www.nass.usda.gov/research/Cropland/sarsfaqs2.html

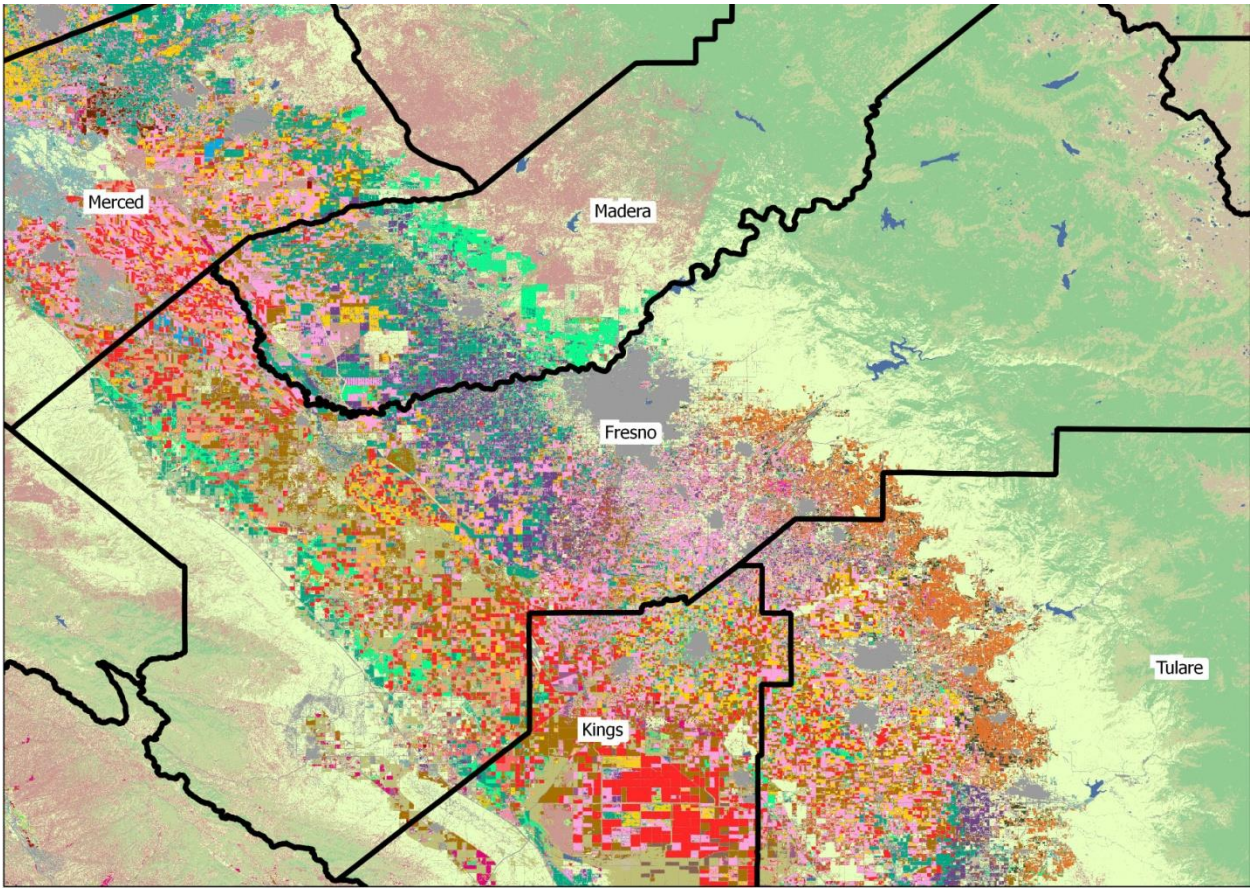


Figure 2.1 – CropScape image for Fresno region in 2010

Presenting the crops corresponding to each color [2] on the map is not reasonable, since there are 256 different colors on the map, but the location of the urbanized area is easily identifiable by the grey area on the center of the map, as well as the cotton production area on Kings County and on the border with Fresno County is also readily identifiable in red.

When analyzing crops only, it was also possible to generate Table 2.1, which presents the most relevant crops (in terms of area) in each county (only the counties with the largest planted areas presented).

[2] - Legend is on the appendix

Table 2.1 – Representativeness of some crops within each county

COUNTY	Crop Areas (1.000 Acres)	Alfalfa	Almonds	Winter Wheat	Rice	Other Hay-Non Alfalfa	Corn	Cotton	Grapes	Walnuts	Oats	Oranges	Pistachios	Tomatoes	Durum Wheat
Fresno	1,107	8%	20%	9%	0%	0%	1%	15%	18%	1%	0%	2%	5%	10%	1%
Tulare	635	13%	9%	7%	0%	1%	4%	5%	9%	6%	3%	16%	5%	0%	0%
Kern	748	11%	27%	7%	0%	1%	2%	10%	11%	0%	1%	4%	10%	2%	0%
San Joaquin	513	14%	11%	8%	1%	3%	14%	0%	11%	14%	4%	0%	0%	8%	0%
Merced	491	21%	24%	6%	0%	3%	5%	13%	1%	1%	4%	0%	2%	5%	0%
Kings	479	12%	6%	15%	0%	0%	3%	29%	2%	3%	1%	0%	5%	6%	1%
Imperial	420	34%	0%	3%	0%	21%	0%	1%	0%	0%	0%	0%	0%	0%	16%
Stanislaus	345	12%	41%	2%	0%	4%	3%	0%	0%	6%	5%	0%	0%	4%	0%
Madera	300	12%	35%	5%	0%	0%	1%	2%	23%	0%	1%	1%	10%	1%	0%
Yolo	279	15%	5%	15%	15%	2%	7%	0%	4%	5%	1%	0%	0%	14%	0%
Colusa	287	4%	16%	6%	55%	0%	2%	0%	0%	5%	0%	0%	0%	5%	0%
Glenn	244	7%	19%	4%	39%	2%	5%	1%	0%	9%	1%	0%	0%	0%	0%
Sutter	240	5%	4%	6%	53%	0%	4%	0%	0%	13%	0%	0%	0%	5%	0%
Butte	221	2%	20%	2%	50%	0%	1%	0%	0%	18%	0%	0%	0%	0%	0%
Sacramento	155	20%	1%	11%	3%	10%	20%	0%	9%	3%	4%	0%	0%	3%	0%
Solano	151	19%	3%	21%	0%	4%	7%	0%	1%	5%	1%	0%	0%	8%	0%
Modoc	136	23%	0%	5%	0%	51%	0%	0%	0%	0%	1%	0%	0%	0%	0%
Siskiyou	174	65%	0%	1%	0%	10%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Riverside	145	27%	0%	18%	0%	2%	2%	10%	7%	0%	1%	0%	0%	0%	1%
Monterey	64	20%	3%	3%	0%	5%	5%	6%	9%	1%	4%	0%	0%	7%	0%

From Table 2.1, it is possible to see that Fresno County, which is the county with the largest crop area in California, has its crop area equally divided among five crops, with almonds, cotton, grapes and tomatoes all accounting for more than 10% of all county crop area each.

This is not the case, however, of most counties, where most of their crop area is assigned to one or two commodities only, like Imperial County, which produces alfalfa in more than a third of its planted area, and if considered also other hay that not alfalfa and durum wheat, other crops are left with less than 30% of the total county crop area.

As a result of concentration in production of a few products in most counties, it should be expected that the seasonality factor of one County would be reasonably different than other Counties which produce different products, and this should result in an interesting impact on the demand matrix when broken down by season.

Table 2.2, in the other hand, presents the products with the largest crop areas in the state, and it is clear that alfalfa, almonds, rice, wheat, grapes and cotton are by far the most important crops in terms of planted area, responding for almost 60% of the whole planted area in 2010.

Table 2.2 – Products with the most planted area in CA in 2010

Product	% of total crop area	% Cumulative
Alfalfa	15%	15%
Almonds	15%	29%
Rice	8%	37%
Winter Wheat	7%	45%
Grapes	7%	52%
Cotton	7%	58%
Other Hay-Non Alfalfa	5%	63%
Walnuts	4%	68%
Tomatoes	4%	72%
Corn	4%	76%
Pistachios	3%	79%
Double Crop Winter Wheat-Corn	3%	82%
Oranges	2%	84%
Oats	2%	85%

2.2 USDA QuickStats

The National Agricultural Statistics Service (NASS), part of the United States Department of Agriculture (USDA), provides a data recovery website called QuickStats.

In this website, USDA makes available two different types of data: Surveys and Agricultural census. Despite the fact that these two data sources have intrinsic different errors and the use of both of them could not be considered ideal to estimate models, there is no strong argument again using them for seasonality analysis.

This data source does NOT cover all crops that CropScape lists as existing in California, but the differences occur in crops of nearly irrelevant products (e.g. mint), so the errors are dismissible.

From this data source it was possible to derive the yield for most products, including different yields for the same product in different regions/counties. The information on Yields that was not available through this data source was either found on the aggregate report for California published by USDA or on independent websites.

It should be noted, however, census data was only available for the years 2002 and 2007, and therefore not ideal in order to build a seasonality analysis for 2010. However, if survey data was not available for any year from 2007 to 2010, corresponding survey data for 2002 was used.

2.3 USDA's California report

The USDA reports annually aggregate statistics about each crop in the state, including harvested area, production, yield, market value and main producing counties for the entire state of California. Even though most of information on this report aggregates the survey data found in QuickStats, the approximate harvesting period is not found anywhere else except in USDA website, and was the most reliable information for harvesting period found in the data gathering step that preceded the development of this report.

It is necessary to say that the USDA report provides approximate dates for start and end of harvest, and it was considered that the harvest would happen according to a triangular distribution during the period comprised by the two dates provided.

This last assumption allows to assign a percentage of each crop in each FAZ to a particular day of the year and, therefore, to a season. Table 2.1 presents the distribution per season for the 18 of the most relevant products.

Table 2.3 – Percentage of each crop harvested in each season of the year

Products	Spring	Summer	Fall	Winter
Alfalfa	25%	50%	25%	0%
Tomatoes	18%	50%	32%	0%
Grapes	17%	44%	40%	0%
Rice	0%	23%	77%	0%
Corn	0%	23%	77%	0%
Winter Wheat	54%	46%	0%	0%
Other Hay-Non Alfalfa	27%	51%	22%	0%
Oranges	30%	24%	16%	30%
Strawberries	49%	51%	0%	0%
Almonds	0%	57%	43%	0%
Walnuts	0%	25%	75%	0%
Olives	0%	0%	51%	50%
Potatoes	25%	26%	25%	25%
Durum Wheat	30%	70%	0%	0%
Pistachios	0%	8%	92%	0%
Oats	5%	86%	9%	0%
Cotton	0%	0%	88%	12%
Barley	26%	67%	7%	0%

3 Relevant products

Even though almost 70 products are listed as being produced in California by the analysis on the CropScape framework, 14 crops account for 85% of all the area harvested in the State in 2010[3], while a different set of also 14 crops account for 85% of the total agricultural production in California3.

The union of these two sets results in a set of 18 products, listed on Table 3.1, which cover about 89% of all the agricultural production of CA and over 87% of the harvested area3.

As the product list presented on Table 3.1 is not too long to suggest a different approach, it was attempted to recover for all of them specific yields per county (or other geographic area smaller than the State) in order to compute a more accurate seasonality estimate.

Table 3.1 – Main agricultural products in California ordered by production

Product	% of total crop area	% of total production
Tomatoes	4%	30%
Alfalfa	15%	15%
Grapes	7%	12%
Rice	8%	5%
Double Crop Winter Wheat-Corn	3%	3%
Corn	4%	3%
Winter Wheat	7%	3%
Sugarbeets	0%	3%
Other Hay-Non Alfalfa	5%	3%
Almonds	15%	2%
Oranges	2%	2%
Onions	1%	2%
Triticale	0%	2%
Carrots	0%	2%
Walnuts	4%	1%
Pistachios	3%	1%
Cotton	7%	1%
Oats	2%	0%

[3] Based on the methodology proposed in this document

Regardless of the effort to collect data for County specific yields, information could only be found for barley, corn, cotton, strawberries, tomatoes and wheat, which combined represent around 43% of the State's annual agricultural production[4].

4 Results

By computing the production of all products for all FAZs in each season, it is possible to compute a distribution of production in each one of these areas. As shown in Figure 4.1, a group of zones north of San Francisco have a very concentrated production, with more than 75% their annual total being produces in a single season.

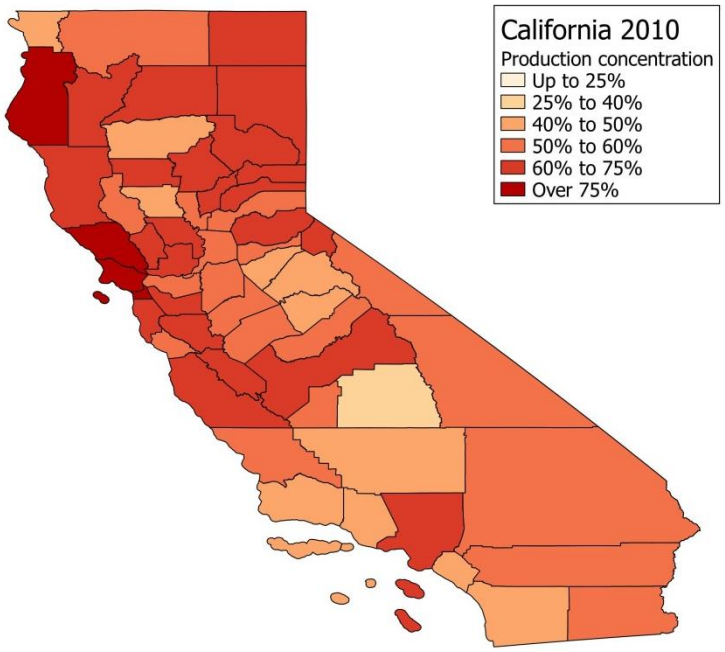


Figure 4.1 – Production concentration in each FAZ

In the other hand, most of the state has a concentration of around 40% to 60% of production in a single season, which corresponds to having roughly half of their production in one season and the rest of it distributed on the other seasons of the year.

The set of maps on Figure 4.2 presents the percentage of production of each FAZ for each season of the year. It is pretty clear that most of the production is distributed on Summer and Fall, but it is also notable that the production in Southern California during Spring is particularly more relevant than the production of Northern California for the same period, which is most likely the result of the harsher winter faced by the northern part of the State.

[4] Based on the methodology proposed in this document

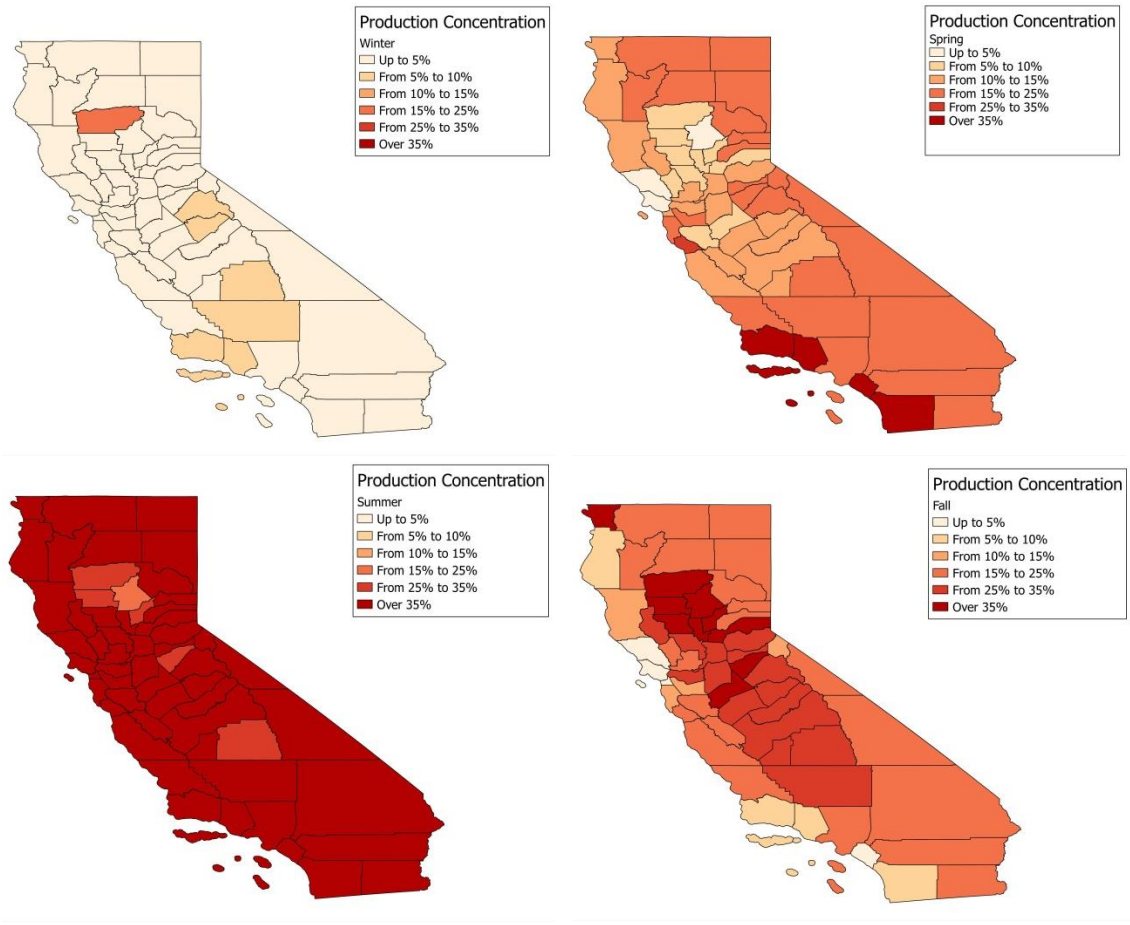


Figure 4.2 – Percentage of annual production for each County in each Season of the year

As a byproduct of the seasonality analysis it is also possible to derive the total production of each County on each season, presented on Figure 4.3.

Although these number are not going to be used in the demand model, they provide an easy way to make a consistency check for the production of a major part of the procedure suggested.

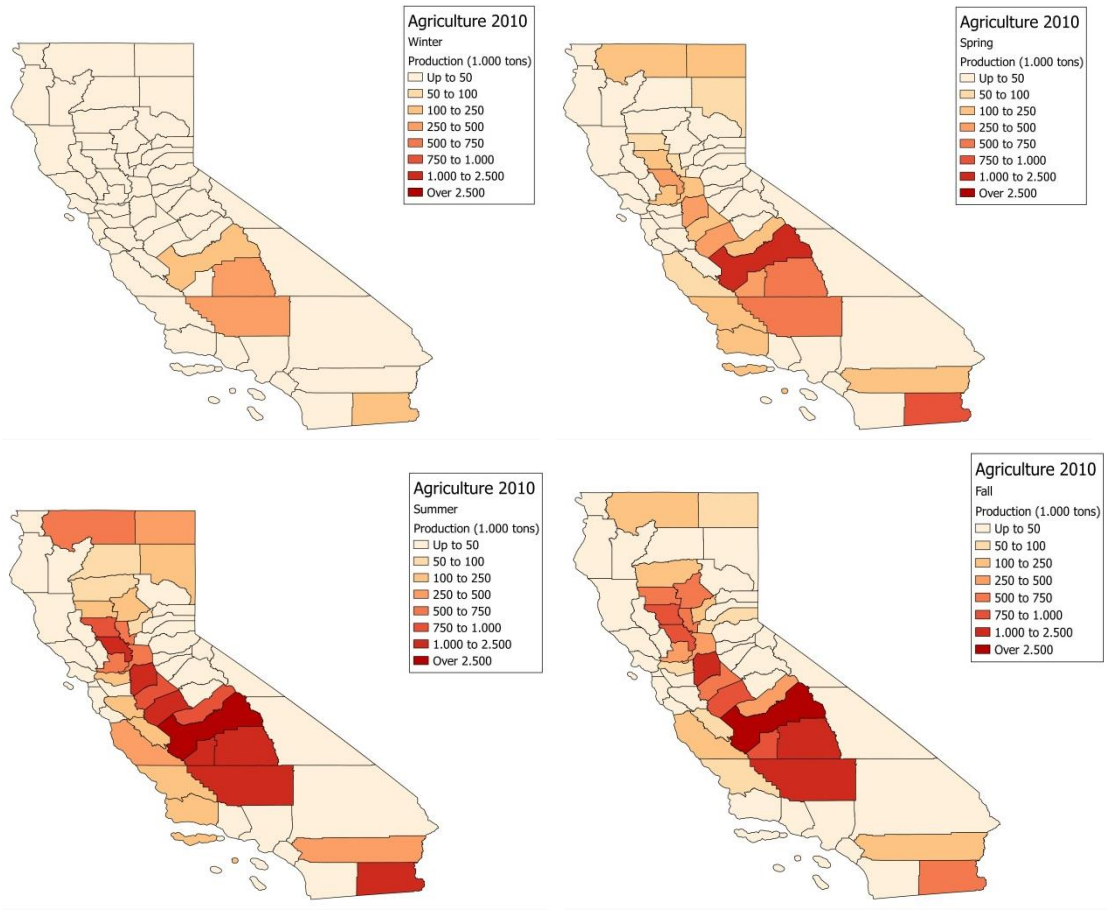


Figure 4.3 – Percentage of annual production for each County in each Season of the year

5 Procedure shortcomings

The most relevant shortcoming of this procedure is the no differentiation between movements from the fields to the retail/wholesale and those from the fields to storage facilities and from there to final consumer, which extends the actual season of those products.

The second most important shortcoming of this procedure is the imprecision of the CropScape map layers that comes to its construction procedure (the image resolution for 2010 was of 56m, or .77 Acre per pixel).

6 Alternate forecasting methodology

Gathering the data for developing this report made clear that there are characteristics of the production process of agricultural goods (crops) that are extremely relevant and have not been included on CSFFM, and neither could, given the

framework established, but that is something that should be reviewed on a next generation of this model.

From an industry stand point, forecasts will most likely be made with regards to planted area, irrigation and expected yield (function of improvement in technology, fertilizer, irrigation ,etc), therefore these variables should be included in the model.

Just as an example, the productivity of irrigated barley in Fresno or Kern Counties is about the double of the productivity of non-irrigated Barley, difference that jumps to over 4 times in the case of Madera county. The case of Wheat is even more extreme, as the irrigated crops have a yield that is larger than the non-irrigated ones by a factor of 3 in Fresno, 2 in Kern and 14 in Imperial county.

7 References

1) CropScape- Cropland data layer, National Agricultural Statistics Service, USDA (http://nassgeodata.gmu.edu/CropScape/)

2) National Agricultural Statistics Service, USDA (http://quickstats.nass.usda.gov/)

3) National Agricultural Statistics Service, USDA (California Report for 2010): http://www.nass.usda.gov/Statistics_by_State/California/Publications/California_Ag_Statistics/index.asp

4) California association of pomegranate producers: http://www.pomegranates.org/index.php?c=7

5) California association of asparagus producers: http://www.calasparagus.com/ConsumerInformation/FAQs.html

6) Website on vegetable gardens: http://www.pickyourown.org/CAharvestcalendar.htm

8 Appendices

8.1 Legend for CropScape maps

